[INCH-POUND]
A-A-3044
November 7, 1996

COMMERCIAL ITEM DESCRIPTION

SANDER, ELECTRIC, PORTABLE, DISK

The General Services Administration has authorized the use of this Commercial Item Description for procurement of type II, class 1 and 2, sanders specified in OO-S-90C for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers a standard duty and heavy duty, portable, electric disk sander.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The disk sander shall be capable of converting to a grinder with appropriate grinding wheel and guard. The disk sander shall consist of housings, electric motor, switch, wiring, gears, spindle shaft, bearings, handles, and standard equipment with a design to provide convenient and safe operation. The sander shall conform to the requirements of Underwriters' Laboratories UL 45 or UL/CSA 745 for electrical tools; safety devices, guards, and ventilating openings shall be within the acceptable safety requirements of UL 45 or UL/CSA 745. Convenient access shall be provided to areas where the manufacturer recommends periodic lubrication or lubricant replacement. All parts subject to wear, breakage, or distortion shall be easily accessible for adjustment and repair. All parts from the same manufacturer shall be interchangeable without modifiable adjustment. The sander shall be similar to Figure 1 and conform to the requirements of Table I.

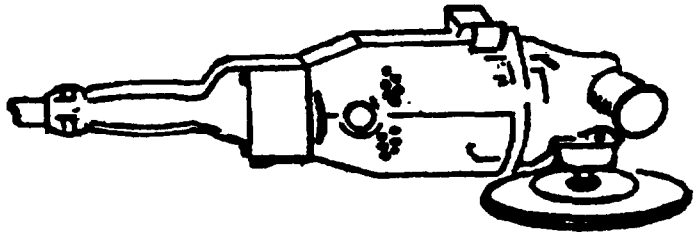


Figure 1. Sander, electrical, portable, disk.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

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TABLE I. Sanders, disk

	Nominal size (Inches)	Disk size (Diameter) (Inches)	Rated HP 1/ (min.)	No-load speed 2/ (rpm)	Length without pad (max.) (Inches)	Net weight without pad or cable (max.) (Pounds)	Spindle thread dim. RH
Standard duty	7	7	1/2	3500-5500	19	18	5/8-11
Heavy duty	7	7	3/4	3800-6000	21	20	5/8-11
Heavy duty	9	9-1/8	3/4	3800-6000	23	20	5/8-11

1/ Horsepower shall be taken at disk end of spindle with speed not less than the full-load minimum rpm.
2/ The full-load speed shall in no case be less than 55 percent of the rated no-load speed.

3.2 Mechanical features. Castings and housings shall be free of defects or mismatching and made of material suitable to withstand the stresses imposed by normal usage. Fastening devices shall be installed with a means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be permanently deformed. All bearings shall be precision quality, commercially available ball or roller bearings permanently lubricated with dust-proof seals. Gears and shafts shall be fully enclosed and of suitable material designed and supported to efficiently transmit imposed loads.

3.2.1 Spindle. The spindle shall be of a suitably hardened alloy or forged steel and shall be supported by at least two bearings. Outer spindle bearings shall be ball, roller, or a combination of ball and roller and designed to withstand the radial thrust loads imposed. A means shall be provided for locking or holding the spindle to facilitate changing abrasive disks and backing pads.

3.2.2 Handles. The sander shall have an end handle and a reversible side handle designed for mounting on either the right or left of the spindle. The longitudinal axis of the handles shall be located to facilitate ease of handling and operation of the sander under service conditions.

3.2.3 Backing pad. When specified (see 7.2), the sander shall be equipped with a semi-flexible pad with clamp nut for backing the abrasive disks. The pad shall be of such material that it will withstand the maximum centrifugal force developed and shall not deteriorate from heat created by sanding operations.

3.2.4 Grinding wheel guard. When specified (see 7.2), the sander shall be equipped with a wheel guard so as to be used as a grinder.

3.2.5 Tool rest. The disk sander shall be equipped with a tool rest of a suitable resilient material fastened to the housing opposite the sanding disk to enable resting the tool without injury to the sander, spindle, and disk assembly. When specified, the heavy duty sander shall have a provision for suspension eyebolt mounting (see 7.2).

3.3 Electrical features. The sander's motor shall be of a universal series wound or compensated series wound designed to operate on 115 volts (± 6 percent), single phase, 25 to 60 cycles, alternating current (AC). The motor armature shall be dynamically balanced to ensure freedom from destructive vibrations. The motor commutator may be of V ring and sheet mica or of molded construction. The mica shall be undercut between the commutator segments. Molded commutators shall be of such material that it will withstand maximum rotational forces and be immune to deformation from temperature encountered in service. The wiring, flexible cord, and attachment plug shall be in accordance with UL 45 or UL/CSA 745, as applicable. The cord shall have an attachment plug cap, and unless otherwise specified (see 7.2), shall have a minimum length of 6 feet measured from point of entry into the sander. The attachment plug cap shall conform to UL 498 requirements for 2-pole, 3-wire grounding devices or 2-wire polarized for double insulated designed tool rated 15 amperes at 125 volts, unless otherwise specified (see 7.2). The sander shall have a safely and conveniently located trigger switch with a current rating of not less than the rated load current of the sander. The switch may have a lock for holding in the "ON" position provided that actuation and turnoff can be accomplished by the same finger(s) that actuate the switch.

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3.4 Handbook of instruction. The manufacturer shall furnish one copy of the regular instruction book and parts list with each sander. The handbook shall include all illustrations, instructions, and necessary data for safe operation, maintenance, adjustment, and repair. The parts list shall include the manufacturer's part number and ordering instructions.

3.5 Load testing. When required to be tested (see 7.2), the sander shall be operated for a minimum of 30 minutes at no-load. During the test, the sander shall be checked for switch operation or any loss of lubricant. After completing the no-load test, the sander shall be operated at the rated load. When operating temperatures are constant, the sander shall be operated at the rated load for a minimum of one hour. The sander shall then be tested for temperature rise, current input, and dielectric strength and insulation resistance in accordance with the applicable requirements of UL 45 or UL/CSA 745. When the rated load test is completed, operational components (including motor bearings and gears) shall be examined for wear, insecure fastening, and evidence of leakage. Failure to meet any of the above tests shall be cause for rejection."

3.6 Additional requirements.

3.6.1 Electromagnetic interference suppression. When specified (see 7.2), the sander shall be electromagnetic interference suppressed in accordance with the requirements of MIL-STD-461.

3.6.2 Fungus proofing. When specified (see 7.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant, or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function would be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of approximately 0.001 inch to the component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.7 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall meet the permanency of marking requirements in UL 45 or UL/CSA 745.

3.8 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

DoD Standardization Documents: Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 1911-5094.

UL 45 or UL/CSA 745 and UL 498: Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(d) If first article samples are required and the number of samples required.

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(e) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(f) What preservation, packing, packaging, and marking of packages are required.
(g) If backing is required (see 3.2.3)
(h) If grinding wheel is required (see 3.2.3).
(i) If an eye bolt mounting is required for the heavy duty sander (see 3.2.4).
(j) Cord length, if different (see 3.3).
(k) Attachment plug conformance to UL 498, if required (see 3.3).
(l) Load testing, if required (see 3.4).
(m) Electromagnetic interference suppression, if required (see 3.6.1).
(n) Moisture and fungus treatment, if required (see 3.6.2).

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the sanders specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	SIZE
5130-00-596-9728	7-inch diameter disk

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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